DESIGN AND IMPLEMENTATION OF A SMART WATER TANK MONITORING AND AUTOMATIC CONTROL SYSTEM USING DIGITAL LOGIC

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Course: **IFT 211 - Digital Logic Design**

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Notation: + means OR, · means AND, and an overbar means NOT.

A. Sensor encoder

The four sensor inputs are S0, S1, S2 and S3.

The two encoded outputs are:

L1 = S2 + S3

L0 = S1 + S3

B. Full tank detector

The tank is full when both encoded outputs are 1.

FULL = L1 . L0

C. Pump control

The pump switches OFF when the tank is full.

PUMP = (FULL)'

D. Level register

The two D flip-flops store the water level.

DR0 = L1 . (RESET)'

DR0 = L0 . (RESET)'

E. Overflow counter enable

The counter operates only when the tank is full and RESET is inactive.

EN = FULL . (RESET)'

F. Overflow counter

The first counter flip-flop:

DQ0 = EN(Q1 + Q'0)

The second counter flip-flop:

DQ1 = EN(Q1 + Q0)

G. Alarm control

The alarm activates when the counter reaches binary 11.

ALARM = Q1 . Q0

H. Seven-segment display

For active-high segment inputs:

Segment	Boolean expression
a	R1 + R'0
b	1
c	R'1 + R'0
d	R1 + R'0
e	R'0
f	R'1 . R'0
g	R1

These equations correspond to displaying decimal digits 0, 1, 2 and 3.

3. Truth tables

Sensor encoder truth table

Assuming one sensor is active at a time:

S0	S1	S2	S3	L1	L0
1	0	0	0	0	0
0	1	0	0	0	1
0	0	1	0	1	0
0	0	0	1	1	1

Pump and full detector truth table

L1	L0	FULL	PUMP
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

Overflow counter truth table

Present Q1	Present Q0	Next Q1	Next Q0
0	0	0	1
0	1	1	0
1	0	1	1
1	1	1	1

This table assumes EN = 1. When EN = 0, the counter returns to 00 on the next active clock edge.

4. Karnaugh maps

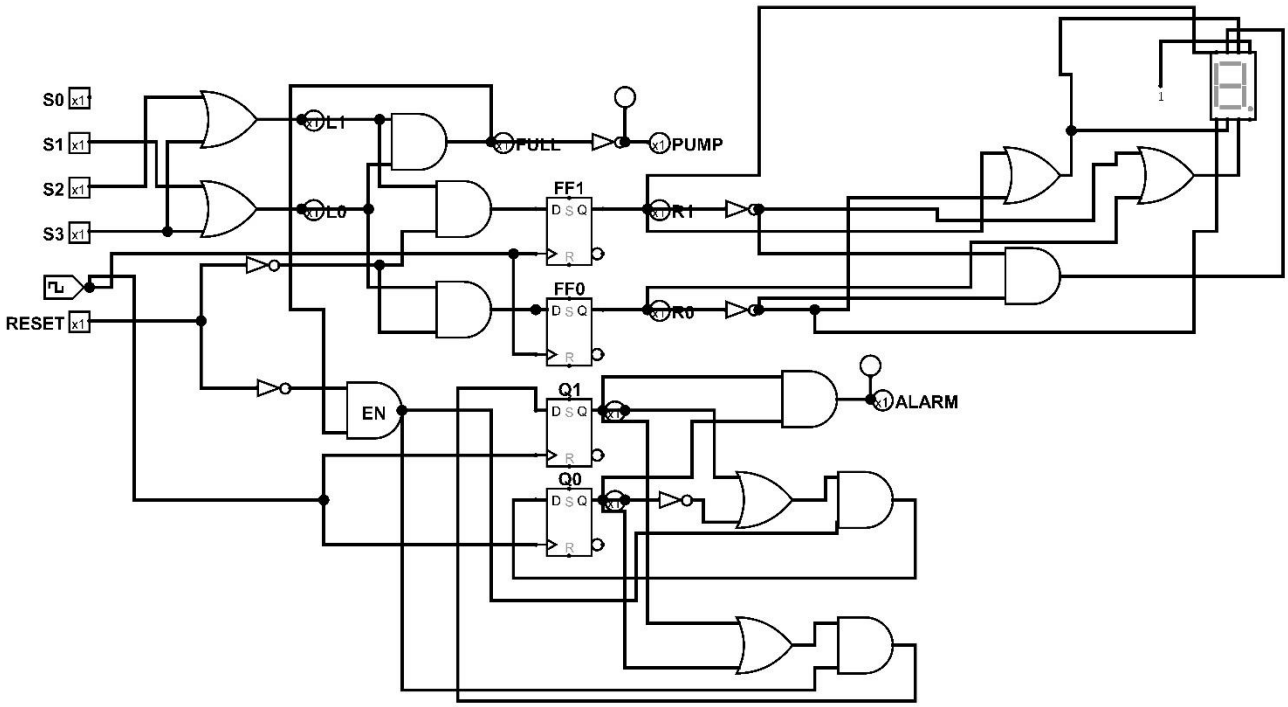
For your report, include K-map simplifications for the main logic functions.

The important simplified equations are:

Function	Simplified expression
L1	S2 + S3
L0	S1 + S3
FULL	L1 . L0
PUMP	(FULL)’
ALARM	Q1 . Q0

5. My Circuit Screenshot/github link

https://github.com/Mr-Ken-Dev/WaterTankLogicCircuit.git



6. Tested scenarios

The circuit was tested in Logisim Evolution using sensor combinations and clock pulses.

The results below are based on the tests reported:

Test	Expected behaviour	Reported result
Seven-segment display	Displays 0, 1, 2, and 3 for the four levels.	Passed
Pump control	ON when FULL=0; OFF when FULL=1.	Passed
Alarm output	ON when counter reaches 11.	Passed
Counter sequence	00 → 01 → 10 → 11 → 11 while enabled.	Passed
Counter reset	Returns to 00 on next rising edge when EN=0.	Passed
Final integration	All circuit sections operate together.	Passed

Counter Test Sequence

Event	Counter state
Initial state	00
First enabled pulse	01
Second enabled pulse	10
Third enabled pulse	11
Further enabled pulse	11 (saturated)
FULL becomes 0; next rising edge	00

7. System Explanation

Four sensor inputs represent the water level. An encoder converts the active sensor into a two-bit binary value. The value is stored in two D flip-flops and displayed. A full detector identifies the full-level code. Its output controls the pump logic and enables the overflow counter. When the counter reaches binary 11, the alarm output becomes active.

NB:

ATTACHED IS A COPY OF MY LOGISIM ON GITHUB

https://github.com/Mr-Ken-Dev/WaterTankLogicCircuit.git